1.3 How do computers represent text, images and sound?

You have already learnt that computers only process binary. This means that all data must be converted to binary to be processed by a computer. You have already looked at converting numbers, but text, images and sound all have to be converted to binary too, so they can be processed by a computer. How is this done?

Converting text to binary

When a computer converts text to binary to be processed it uses a tool called a **character set**. There are two main types of character set that a computer can use, ASCII and Unicode. An ASCII character set represents characters using 8-bit binary numbers. This means that it can represent up to 256 characters (0 to 255). This is fine for certain languages, but some languages have an alphabet that requires more than 256 characters; therefore, Unicode was invented. A Unicode character set represents characters using 16-bit binary numbers. This means it can represent a much greater number of characters, approximately 65 000. This means that many more characters can be developed in different alphabets and also symbols such as emojis can also be used.

Each character has a set binary value, for example, the letter A is 01000001.

KEY WORD

character set: it contains all the characters that are in that character set and the binary value that is assigned to each character.

ACTIVITY 1.2

Use the internet to find an example of an ASCII character set and a Unicode character set. What kind of different characters are represented? Are upper and lowercase letters the same?

Using the ASCII table, try to decode this message:

01000011011011110100110101110000011101010111010001100101011100100101001101

10001101101001010001010110111001100011011001010110100101110011011001100111

01010110111000100001

Remember, each letter will be represented as an 8-bit binary number.

Try writing a message for your friend to decode.

Converting images to binary

An image is made up of tiny dots called **pixels**. If an image was simply created using the colours black and white, each pixel would be either black or white. The binary value 1 could be used to represent the colour black and the binary value 0 could be used to represent the colour white.

KEY WORD

pixel: a very small dot of colour that is displayed with many others to create an image.

If each pixel is converted to its binary value, a data set such as the following could be created:

0011111000100000101000000011001010011000000011010001011001110010100000100
01111100

If the computer is informed that the image that should be created using this data is 9 pixels wide and 10 pixels high, it can set each pixel to black or white and create the image (See Figure 1.2). The type of data that is used to provide information, such as the dimensions and **resolution** of an image, is called **metadata**.

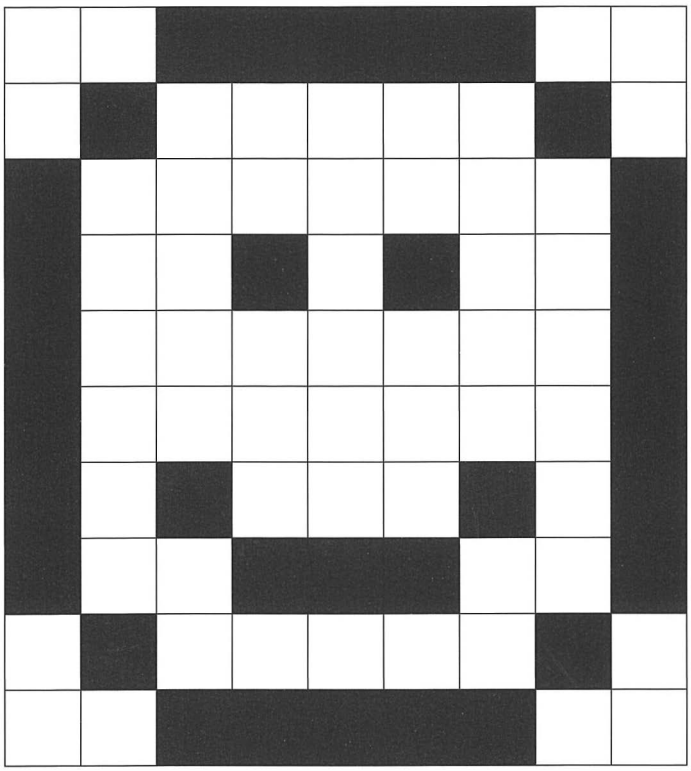


Figure 1.2: An image created just from black and white pixels

Most images use a lot more colours than black and white. Each colour has its own binary values. Colours are created by computer screens using the Red Green Blue (RGB) colour system. This system mixes the colours red, green and blue in different amounts to achieve each colour.

ACTIVITY 1.3

Use the internet to find an RGB colour scale. You should notice that each colour in the scale is first presented as hexadecimal. The first two hexadecimal values are the amount of red, the second two the amount of blue and the last two the amount of green.

Practise your hexadecimal to binary conversion by converting your favourite colour to its binary value.

KEY WORDS

resolution: the dimensions of an image.

metadata: additional data that is stored with an image that can provide information such as the dimensions of the image and the time and date the image was taken.

Most images are created using thousands of pixels. Each of those pixels needs quite a lot of data. Multiply that by the thousands of pixels that make up the image and that is a huge amount of data. This is why image files can often be so large in size.

Each image has a resolution and a **colour depth**. The resolution is the number of pixels wide and the number of pixels high that are used to create the image. The colour depth is the number of bits that are used to represent each colour. For example, each colour could be represented using 8-bit, 16-bit or 32-bit binary numbers (see Figure 1.3). The greater the number of bits, the greater the range of colours that can be represented.

KEY WORD

colour depth: the number of bits that are used to create each colour in an image

If the colour depth of an image is reduced, the quality of the image is often reduced.



Figure 1.3: 8-bit colour 16-bit colour 32-bit colour

Can you see how the quality of the image is reduced in comparison to the number of bits used?

If the resolution or the colour depth of an image is changed, this will have an effect on the size of the image file. If the resolution is increased, the image will be created using more pixels, so more data will need to be stored. If the colour depth of the image is increased, each pixel will need more data to display a greater range of colours so more data will need to be stored. Both will result in a larger file size for the image.

COMPUTER SCIENCE IN CONTEXT

People often have to sacrifice the quality of their images when displaying them online. This is because having files that are very large in size can take more time to download. This may frustrate users as it may mean they have to wait for a website to load, due to the number of large files it contains.

Converting sound to binary

Sound is made up of sound waves. When sound is recorded, this is done at set time intervals. This process is known as **sound sampling**.

KEY WORD

sound sampling: a little piece of sound that is recorded at regular time intervals.

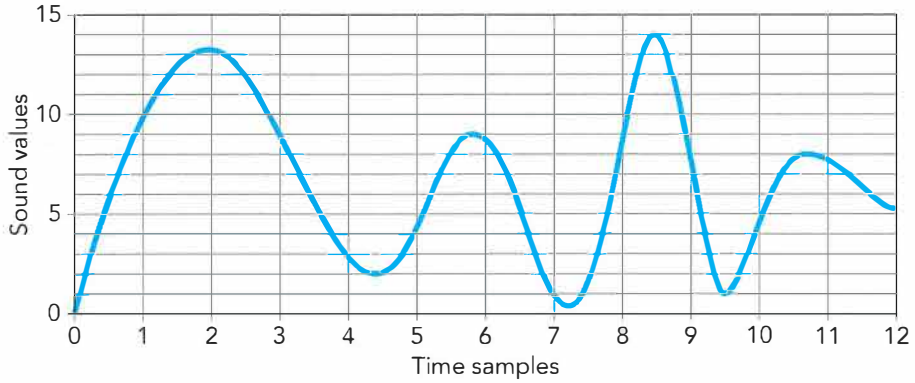


Figure 1.4: A sound wave

Figure 1.4 is an example of a sound wave.

If you recorded the sound at each time sample, you would have the following set of values:

Time sample	1	2	3	4	5	6	7	8	9	10	11	12
Sound value	9	13	9	3.5	4	9	1.5	9	8	5	8	5.5

Each sound value is converted to binary to be processed by the computer. Each sound value is then used to playback the sound, when the recording is played. If the recorded sound wave is played it would now look like Figure 1.5:

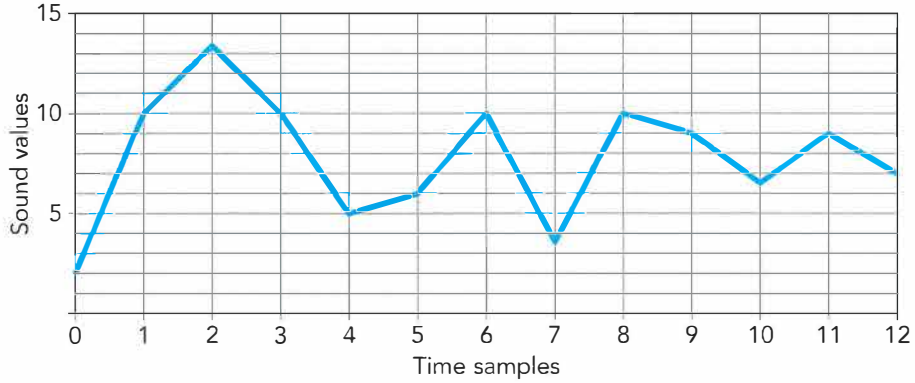


Figure 1.5: A sound wave created by playing back a recorded sound

You will notice that the two sound waves look different. This is because the number of samples taken is too far apart, so vital data in between time samples has been lost. This will result in a lack of accuracy in the sound that is recorded. Therefore, more samples need to be taken each second to improve the accuracy of the recorded sound. The number of samples taken in a second is called the **sample rate**. Sample rates are measured in hertz. 1 hertz is equal to 1 sample per second. A common sample rate is 44.1 khz (kilohertz), this would require 44 100 samples to be taken each second. That is a lot of data! If the sample rate is increased, the amount of data required for the recording is increased. This increases the size of the file that stores the sound.

Further data is required to represent sound, and this is the **sample resolution**. This is the number of bits that are used to represent each sample. A common sample resolution is

KEY WORDS

sample rate: the number of samples recorded each second.

sample resolution: the number of bits that are used to record each sound sample.

16-bit. The higher the sample resolution, the greater the variations in amplitude that can be stored for each sample. This means that aspects such as the loudness of the sound can be recorded more accurately. This will also increase the amount of data that needs to be stored for each sample. This increases the size of the file that stores the sound.

1.4 Measuring data storage

Earlier in the chapter, you learnt that all data need to be converted to digital data to be processed by a computer, and you now know how that is done for different types of data. But how is data measured when it is stored?

COMPUTER SCIENCE IN CONTEXT

There are times when you will need to know the size of a data file, such as when you are attaching a file to an email. Most email providers have a file size limit, so you need to know the size of that file, to know if it can be attached to your email.

Data is measured using different units, which are shown in Table 1.2.

Unit	Description
Bit	This is the smallest measurement for data
Nibble	There are 4 bits in a nibble
Byte	There are 8 bits in a byte
Kibibyte (KiB)	There are 1024 bytes in a Kibibyte
Mebibyte (MiB)	There are 1024 Kibibytes in a Mebibyte
Gibibyte (GiB)	There are 1024 Mebibytes in a Gibibyte
Tebibyte (TiB)	There are 1024 Gibibyte in a Tebibyte
Pebibyte (PiB)	There are 1024 Tebibytes in a Pebibyte
Exbibyte (EiB)	There are 1024 Pebibytes in an Exbibyte

Table 1.2: Data measuring units

Although most computers will calculate the size of a file for you, you need to be able to show that you can do this manually for the exam. This is just for an image file and a sound file.

SKILLS FOCUS 1.3

CALCULATING THE SIZE OF A FILE

There are several pieces of information that you need to know to calculate the size of an image file, these are:

- the number of images in the file
- the resolution of the image
- the colour depth of the image.

CONTINUED

You will be asked to calculate the size of a sound file to a certain unit of measurement. The answer given (10 584 000) is the number of bits that are in the sound file. If you wanted to know how many bytes are in the sound file, you need to divide the final answer again by 8, as there are 8 bits in each byte, for example:

$$10\ 584\ 000 \div 8 = 1\ 323\ 000 \text{ bytes}$$

If you wanted to know how many mebibytes the file was, you need to divide the answer in bytes by 1024, then by 1024 again, for example:

$$1\ 323\ 000 \div 1024 \div 1024 = 1.3 \text{ MiB (to 1 decimal place)}$$

1.5 Data compression

You may already have experience of having large files and struggling to have enough storage space to store them. One thing that can be done to reduce the size of a file is **compression**.

Compressing a file will reduce the size of the file that can have several benefits, such as:

- Not as much storage space is needed to store the file.
- It will take less time to transmit the file from one device to another.
- It will be quicker to upload and download the file.
- Not as much bandwidth is needed to transmit the file over the internet.

There two type of compression that can be used, **lossy** and **lossless**. Each compresses the data in a different way.

Lossy compression uses a compression algorithm that finds the unnecessary and redundant data in the file. This data is permanently removed from the file. This type of compression is mainly used on an image file or a sound file.

Unnecessary data in an image file that could be removed are colours that the human eye cannot distinguish. This means that a lossy compression algorithm for an image file could reduce the size of the file by reducing the colour depth. It could also reduce the size of the file by reducing the image resolution, reducing the number of pixels used to create the image.

Unnecessary data in a sound file that could be removed are sounds that cannot be heard by the human ear. It could also be the removal of softer sounds that are played at the same time as louder sounds, as humans will only hear the louder sound. The method that is used to find these kinds of sounds in a sound file is called **perceptual music shaping**. This is the most common type of method that is used in the file format MP3. The size of a sound file can also be reduced by reducing the sample rate and the sample resolution.

KEY WORDS

compression: a method that uses an algorithm to reduce the size of a file.

lossy: a compression method that reduces the size of a file by permanently removing data.

lossless: a compression method that reduces the size of a file by temporarily altering the data.

KEY WORD

perceptual music shaping: a process that is used in lossy compression that removes sounds that are not normally heard by the human ear.

Questions

7 Why do you think lossy compression is not commonly used on text files?

8 Will lossy compression affect the quality of an image? Why?

9 Will lossy compression affect the quality of a sound recording? Why?

SELF ASSESSMENT

What were the points that you thought about regarding lossy compression to answer question 7? Did you relate these points back to why lossy compression is not suitable for a text file, or did you just make points about the drawbacks of lossy compression in general?

Lossless compression uses a compression algorithm to reduce the size of the file, without permanently removing any data. A common method that can be used to do this is by finding repeating data in the file and grouping this data together. For example, consider the following message:

WHEN IT IS SNOWING HEAVILY LOOK OUTSIDE.
LOOK OUTSIDE IT IS SNOWING HEAVILY.

Excluding the spaces between the words and the full stops, the message has a total of 62 characters. 1 character requires 1 byte of storage, so 62 bytes of storage would be needed to store this message.

When you look at the message, you should notice that it consists of words that are mostly repeated. Instead of storing all 62 characters individually, the words and the positions at which they occur in the message can be stored in a lookup table, for example:

Word	Position(s) in the message
WHEN	1
IT	2, 10
IS	3, 11
SNOWING	4, 12
HEAVILY	5, 13
LOOK	6, 8
OUTSIDE	7, 9

The lookup table needs 1 byte for each character in each word and 1 byte for each position the word occurs in the message. Therefore, 33 bytes are needed to store the words and 13 bytes to store the positions, giving a total of 46 bytes. This is much less than the 62 bytes we required with our original method. No data has been lost and we have reduced our storage requirements by 26%, quite a saving! To recreate the message, the computer simply retrieves the words and places them in the positions allocated.

A similar process can be applied to an image file with a lossless compression algorithm called **run length encoding (RLE)**. This algorithm will group together repeating pixels and store how many times they occur. For example, consider the simple image in Figure 1.6:

KEY WORD

run length encoding (RLE): An algorithm that groups together repeating patterns and indexes them.

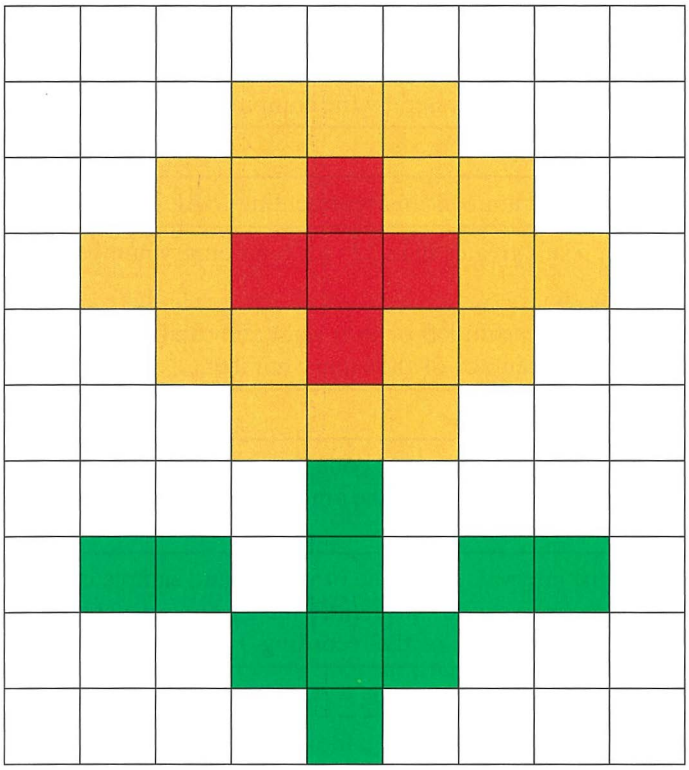


Figure 1.6: A simple pixel image

If W is white, Y is yellow, R is red and G is green, RLE could compress the image into the following data:

12W, 3Y, 5W, 2Y, 1R, 2Y, 3W, 2Y, 3R, 2Y, 3W, 2Y, 1R, 2Y, 5W, 3Y, 7W, 1G, 5W, 2G, 1W, 1G, 1W, 2G, 4W, 3G, 7W, 1G, 4W.

This means that the amount of data can be reduced as pixels are grouped together and the data for each pixel does not need to be stored. The resolution of the image is stored with the data, so that the computer knows the size to recreate it when the data is decompressed.

ACTIVITY 1.4

Why don't you try creating a simple image and writing out the RLE data for it? You could then give it to your friend, along with the image resolution, and they can have fun trying to recreate your image.

Peer Assessment

Ask your friend whether you got the RLE correct for your image in Activity 1.4. Ask them if you could have done anything to make your RLE any clearer. Ask them if you missed any important data.